Hospital Prototype Scorecard Spreadsheet Review

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On July 30, 2026, the Summary of Major Comments section was updated to follow the structure of four groups that later reviews contained. Changes shown in square brackets [].

Summary of Major Comments

The following comments represent errors found when comparing values from the scorecard to the source cited in the scorecard:

- Wall U-value for 1978-1997 vintage (HOSP22-Envelope)
- Parking area lighting power (HOSP45-Ext Lights)

The following are the highest priority comments:

- The schedules show zero occupancy at night (HOSP54-Schedules)
- Single zone basement as office is not typical (HOSP13-Zones)
- Too many occupants assumed in corridors, nurses' stations and lobbies (HOSP20-Zones)
- Number of assumed beds seems small and the building seems small (HOSP9-Prototype, HOSP10-Prototype)
- Some space types overrepresented and some underrepresented (HOSP17-Zones)
- Laundry and special sanitation space types missing which lowers water heater load (HOSP36-Water Heater)

The following comments represent other important issues found:

- Real hospitals are seldom simple boxes (HOSP5-Prototype)
- No shading from entrance overhang (HOSP6-Prototype)
- Low window to wall ratio (HOSP7-Prototype)
- Many spaces-types in hospital not represented (HOSP15-Zones)
- Not enough HVAC Zones for complexity of hospital (HOSP16-Zones)
- No heat recovery is used (HOSP25-HVAC Systems)
- No thermostat setback used for office portions (HOSP32-Thermostat)

- Not enough special space-types in hospitals which have high ventilation or unusual control requirements (HOSP35-Ventilation)
- Water heater sizing not sensitive to water main temperature (HOSP39-Water Heater)
- Additional lighting power that is allowed is not used (HOSP40-Interior Lights)
- No shading assumed but exterior lighting for canopies used (HOSP49-Ext Lights)
- More substantiation needed for operating room and imaging room equipment power densities (HOSP51-Equipment, HOSP52-Equipment)

In addition, the following comments describe the lack of disclosed assumptions that make verifying the values in the scorecard nearly impossible:

- None

Background

The review of the "score card," which is a spreadsheet summarizing the model assumptions, is broken down by spreadsheet tab, which includes:

- Notes
- Prototype
- Zones
- Envelope
- HVAC System
- Thermostat
- Ventilation
- Water Heater
- Interior Lighting
- Ext Lighting
- Equipment
- Schedules
- Energy Usage
- Weights
- PV
- EV Charger

The comments on these tabs are labeled using the following format HOSPx-<TAB>-<CELLRANGE>), where <TAB> is the name of the tab in the spreadsheet and

<CELLRANGE> indicates the specific cells. Note that no tabs were reviewed after Schedules.

HOSP1-Notes (n/a)

Rather than repeat it, please see comment PPD1 in my single-family residential prototype review, LHSC1-Notes in the large hotel review, and SHSC1-Notes in the small hotel review.

HOSP2-Notes (n/a)

Rather than repeat it, please see comment PPD2 in my single-family residential prototype review, LHSC2-Notes in the large hotel review, and SHSC2-Notes in the small hotel review.

HOSP3-Notes (A1:C24)

Due to the complexity of hospitals, the modeling of them is almost a subspecialty of building energy modeling. To provide a thorough review of the prototype, I reviewed several great resources shown below prior to beginning my review. Originally, almost 80 possible resources were identified, but the following seemed the most important.

Sources that *should* be reviewed for background information and are fundamental for reviewing the hospital prototype are the following:

- *ASHRAE Handbook HVAC Applications Chapter 9 "Health Care Facilities"*
  Essential for understanding healthcare HVAC. If only one thing is reviewed, this should be it.
  https://store.accuristech.com/standards/2023-ashrae-handbook-hvac-applications-i-p?product_id=2225673
- *Large Hospital 50% Energy Savings: Technical Support Document*
  Lots of useful details for hospital spaces and HVAC. Contains mapping between space types categories in Appendix A. Schedules in Appendix E.
  https://docs.nrel.gov/docs/fy10osti/47867.pdf
- *ANSI/ASHRAE/ASHE Standard 170-2021 Ventilation of Health Care Facilities*
  Contains a table of design parameters related to ventilation
  https://store.accuristech.com/standards/ashrae-170-2021?product_id=2212971
- *Facility Guideline Institute's Guidelines for Design and Construction of Hospitals 2018*. So many details on specific areas of hospitals. Almost too much information.

A newer version is available that was not reviewed.
https://fgiguidelines.org/

- *ASHRAE's HVAC Design Manual for Hospitals and Clinics, Second Edition*
  Essential HVAC-oriented book on health care design.
  https://store.accuristech.com/standards/hvac-design-manual-for-hospitals-and-clinics-2nd-ed?product_id=1852624
- *Building Type Basics for Healthcare Facilities 2nd Ed*
  Describes layouts and arrangements of hospitals, typical areas, and the length and width of rooms, some information on diagnostic rooms, and their use. Essential reference.
  https://www.bnibooks.com/products/building-type-basics-for-healthcare-facilities-second-edition?_pos=2&_sid=92a3e800a&_ss=r
- *Targeting 100! Envisioning the High Performance Hospital: Implications for a New, Low Energy, High Performance Prototype*
  Includes two forms of hospital prototypes that include code baseline as well as high efficiency options. It was a great source of information since it was focused on developing a prototype for hospitals. The website has lots of supplementary information that is worth reviewing, including: architectural basis, multiple mechanical bases, modeling results, and modeling assumptions.
  https://t100idl.be.uw.edu/t100-library/
- *Hospital Building and Departmental Area Calculation: Comparison of 36 Recent North American Projects - Lavy*
  Provides information on space usage in a building by department.
  https://pubmed.ncbi.nlm.nih.gov/31446796/

Additional useful sources could be added to the list shown in the scorecard spreadsheet:

- *Better design, better hospitals*
  Good background on evidence-based design, which is driving many hospital designs
  https://pmc.ncbi.nlm.nih.gov/articles/PMC5008943/
- *ASHRAE's Advanced Energy Design Guide for Large Hospitals – 50%*
  TSD is more important
  https://www.ashrae.org/technical-resources/aedgs/50-percent-aedg-free-download
- *Healthcare Facility Planning: Thinking Strategically - Second Edition - Hayward*
  Useful space planning book with examples and some typical numbers for area and

counts
https://www.ache.org/learning-center/publications/books/2307i

- *ModelKit Caboodle - healthcare-hospital prototype*
  ModelKit hospital prototype.
  https://bigladdersoftware.com/
- *Producer Price Index for New Health Care Building Construction Sector, NAICS Statistics on construction*
  https://www.bls.gov/ppi/factsheets/producer-price-index-data-for-nonresidential-building-construction-sector-new-health-care-building-construction-naics-236224.htm
- *Northwest Energy Efficiency Alliance – Commercial Building Stock Assessment 2019*
  https://neea.org/commercial-building-stock-assessment/
- *Natural Resources Canada – Survey of Commercial and Institutional Energy Use 2019*
  https://oee.nrcan.gc.ca/corporate/statistics/neud/dpa/menus/scieu/2019/tables.cfm

Some additional data sources have been identified, but were not reviewed. The reasons that they were not reviewed were due to cost, the rawness of the data, or the time available:

- *American Hospital Directory*
  https://www.ahd.com/
- *Architecture and Health Guiding Principles for Practice - Battisto, Dina, Wilhelm*
  https://www.routledge.com/Architecture-and-Health-Guiding-Principles-for-Practice/Battisto-Wilhelm/p/book/9780367075224
- *SpaceMed Hospital and Healthcare Facility Planning Guidelines*
  Looks very useful, but costs $545
  https://www.spacemed.com/
- *State of U.S. Health Care Facility Infrastructure - ASHE*
  Might be useful – hard to know. Available to members
  https://www.ashe.org/facilityinfrastructure
- *AHA Hospital Statistics 2025 Edition*
  Costs $349 and hard to know if it would be useful
  https://ams.aha.org/EWEB/DynamicPage.aspx?WebCode=ProdDetailAdd&ivd_prc_prd_key=a59b2c68-6db8-4bf5-b67c-8418a1fc5a26

- *California Department of Health Care Access and Information (HCAI) Facility Detail*, which includes hospitals in California, is a database that could be mined https://ams.aha.org/EWEB/DynamicPage.aspx?WebCode=ProdDetailAdd&ivd_prc_prd_key=a59b2c68-6db8-4bf5-b67c-8418a1fc5a26
- *Hospital Building Data - Dataset - California Health and Human Services (CalHHS)* https://data.chhs.ca.gov/dataset/hospital-building-data https://data.chhs.ca.gov/dataset/hospital-building-data/resource/d97adf28-ebaf-4204-a29e-bb6bdb7f96b9
- *Hospital Building Data (HHS Open Data)* https://healthdata.gov/State/Hospital-Building-Data/txjx-h3qx/about_data Maybe useful for the number of stories.

By utilizing additional sources of information, the validity of assumptions can be strengthened. Some parameters may be California-specific, but comparing assumptions across multiple databases lends credibility to any assumptions made, even when a California-specific database ultimately may be the source of the information. Tables showing parameters across multiple databases, and then values chosen being highlighted, can be a powerful expression of when parameters are solidly determined by being similar in multiple data sets versus being based on just a single data set. Please also note that some of the above references may include citations of relevant reports and papers that could be reviewed as further sources.

HOSP4-Notes (C2:C5, C11:C19)

Provide links for each reference. Also note that C19 contains a link that should probably be in C18.

HOSP5-Prototype (A7:C7, A9:C9)

The shape of the hospital, being a simple rectangle, seems very unlikely. While I have not visited a random sampling of hospitals, I can say that none were simple rectangles. This will result in decreased skin loads compared to other loads in the building. This is exasperated by a relatively square (aspect ratio of 1.3) rectangle. The Target 100 prototypes looked like:

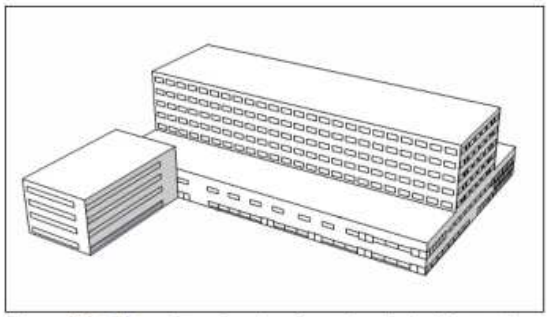


Figure 3.0: Three dimensional configuration of Architectural Scheme A: the Typical form

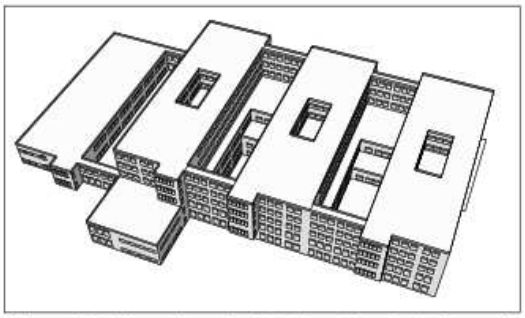
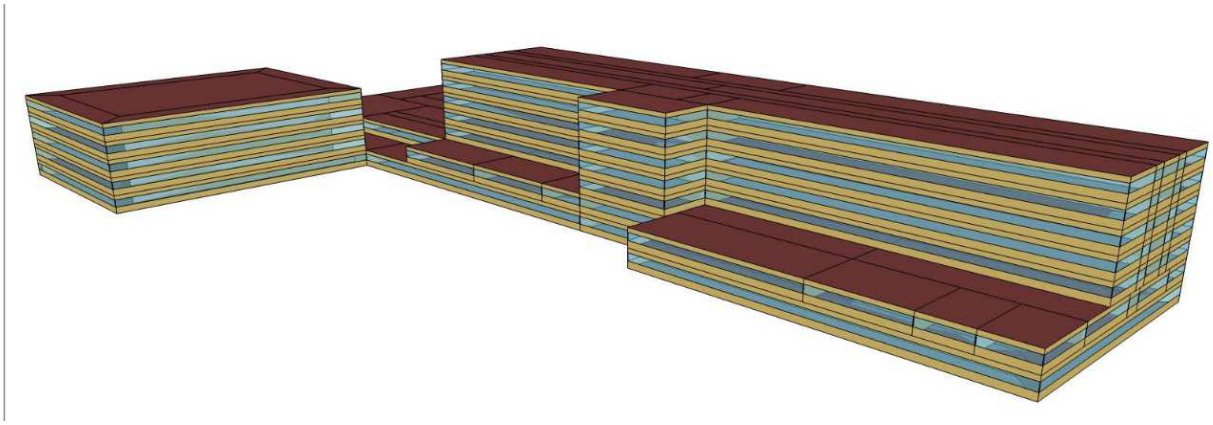


Figure 4.0: Three dimensional configuration of Architectural Scheme B: the Perforated form

The AEDG 50% hospital looked like:



HOSP6-Prototype (A13:C13)

The model shows no shading geometry; this is very unlikely for the public entrances to a hospital, which almost always have overhangs over the entrance to protect visitors from rain when being dropped off. The Target 100 prototype had 21,949 sqft of shaded windows of 35,216 sqft of total window area, about 62% of the fenestration was shaded.

HOSP7-Prototype (A12:C12)

The window-to-wall ratio of 16% seems low to me based on the prevalence of evidence-based design and its focus on the health benefits of connecting patients to views of nature. There are many types of spaces in a hospital that are unlikely to have windows, and those are usually put in the core of the building. The façades of hospitals generally have a lot of fenestration (see the AEDG prototype rendering in comment HOSP5-Prototype) at 40% WWR. The Target 100 prototype assumed a 29% window-to-wall ratio for the baseline and 31% for the high performance prototype.

HOSP8-Prototype (A12:C12)

It would be good to state the window-to-wall ratio by categories of space types. Patient rooms and offices would be of special interest.

HOSP9-Prototype (n/a)

The number of beds is a metric frequently used for hospital size and should be included in the summary. Based on the number of people shown in "Patient Rooms" on the "Zones" tab, this should be 102. According to the AHA Hospital Statistics 2019, the U.S. had 6090 hospitals and 919,559 beds, which averages to 150 beds per hospital. That source also shows:

CLASSIFICATION	HOSPITALS	BEDS
UNITED STATES	6,090	919,559
6-24 Beds	842	13,744
25-49	1,396	46,053
50-99	1,160	82,269
100-199	1,195	170,099
200-299	619	150,576
300-399	372	129,264
400-499	179	79,535
500 or more	327	248,019

The largest number of beds is actually in the 500 or more category, which averages 758 beds per hospital. Assuming we use the 150 beds/hospital, that is significantly larger than the 102 beds/hospital assumed in the prototype. It would also be good to report the average size of the patient rooms and the number of patients in each room (a typical modern design is one patient per room).

HOSP10-Prototype (A8:C8)

The floor area assumed, 241,500 square feet, is based on the PNNL prototype and based on 102 beds (see previous comment), which comes to 2,368 sqft/bed. This is clearly a rough metric that is not related to the size of the patient rooms, but rather to the overall capacity of the hospital in all of its facilities. The Targeting 100 work for the two prototypes used 476,094 sqft or 477,601 sqft each with 226 beds, which is 2,107 sqft/bed to 2,113 sqft/bed. The "Hospital Building and Department Area Calculation" paper by Lavey of 36

recent North American hospitals showed an average of 410,246 sqft and an average of 153.1 patient beds, which comes to 2,683 sqft/bed. This further supports using a larger prototype. The AEDG 50% model has 427,000 sqft (excluding the medical office building), but the number of beds is not clear.

HOSP11-Prototype (A8:C8)

The floor-to-floor height is shown as 8 feet for the ground floor and 14 feet for the upper floors, but the floor-to-ceiling height is shown as "No ceiling; plenum is modeled". This statement is unclear. Is there no ceiling, but the plenum is modeled? That seems like a contradiction. If the semicolon is removed, then it would be more self-consistent: "No ceiling plenum is modeled." Please explain if a plenum space is modeled and used as a plenum space in the model. In general, for a building with such high ventilation rates, such as a hospital, the impact of plenums is greater and should be modeled.

HOSP12-Prototype (A19:B22)

Some of the calculated areas are incorrect:

Shown:

Above grade wall area (ft²)	56,720
Gross wall area (ft²)	63,203
Roof area (ft²)	40,254
Underground Gross Wall	6,482

Recomputed:

- Above grade wall area (ft^2) = 51840 = (4 x 14 + 8) * (230 + 175) * 2
- Gross wall area (ft^2) = 58320 = 51840 + 6480
- Roof area (ft^2) = 40250 = 230 * 175
- Underground Gross Wall Area (ft^2) = 6480 = 8 * (230 +175) * 2

The last two values are probably just based on rounding from SI numbers. The first computation for the above grade wall area is quite far off. It seems to be based on all floors being 14 ft, but B14 says the ground floor is eight ft. Perhaps this is intended to say the basement floor height is 8 ft, and all the above-ground floors are 14 ft. It is not clear if these numbers are ever used, even indirectly, in the model.

HOSP13-Zones (A3:L3)

Like the large hotel, this prototype model also has a single-zone basement shown as an "office area" that has over 40,000 sqft and contains 201 people. This is not the only office space in the hospital; it contains 2,550 sqft distributed on floors 1 through 4 (A30:L34). This large basement office comprises 16.7% of the total floorspace in the hospital. A basement in a hospital probably has many purposes that do not require windows or require putting heavy equipment directly on the foundation, including imaging, laboratories, refrigerated rooms, secure storage, laundry, mechanical, etc. It is unlikely that an office would take up the space since administration is much more likely to be located with windows. In addition, it is not clear what these 201 people in the basement would possibly be doing for a hospital with 102 patient rooms. The basement should be rezoned to include specific spaces that make sense for it.

HOSP14-Zones (E2:E57)

The area fraction column often shows zero since it is formatted as a percentage with no digits after the decimal place. This makes smaller zones show as 0%. It should be formatted to show one digit after the decimal place.

HOSP15-Zones (A2:A58)

All zones in the hospital prototype fall into one of the space types:

- Corridor Area
- Dining Area (Cafeteria/Fast Food)
- Exam/Treatment Room
- Imaging Room
- Nurse's Station
- Operating Room
- Patient Room
- Physical Therapy Room
- Recovery Room
- Kitchen/Food Preparation Area
- Laboratory-General Health
- Lobby, Main Entry
- Office Area

But this lacks many types of spaces often seen in hospitals. Table 3-2 "Space Type Floor Ares in the Prototype Model" from the AEDG 50% TSD shows 32 different space types. Several in that table could be seen as more detailed sub-types of those listed above, but some are shown in Table 3-2 that have no corresponding item, those include:

- Anesthesia gas storage
- Clean workroom/holding
- Laundry
- Locker
- Mechanical/electrical/telecom
- Medical Supply
- Pharmacy
- Restroom
- Soiled workroom/holding
- Sterilizer equipment room
- Storage/receiving

And some space types that may not map directly, like:

- Nursery
- Trauma/emergency room
- Triage

Standard 170 includes Table 7-1 Design Parameters-Inpatient Spaces, which by itself has dozens of space types, many of which could be mapped to the list currently in the prototype. The Targeting 100 Final report includes a program summary in Appendix 1 that includes almost 50 different space types. The following space types shown in that report that are missing from the hospital prototype and do not appear to be sub-types are:

- Pathology
- Inpatient pharmacy
- Sterile Processing
- Chapel/Meditation
- Provider Sleep Rooms
- Admitting
- Medical Records
- Gift shop
- Biomedical Engineering
- Linen Services

- Materials Management
- Loading Dock
- Environmental Services
- Bed Repair and Storage
- Staff Amenities
- Communications
- Building Main Telecom
- Storage
- Mechanical

And some space types that may not map directly, like:

- Labor & Delivery
- Newborn Nursery
- NICU Level II
- Respiratory Services

Every building energy modeler needs to decide which types of spaces to lump together. The combining of spaces should be done with thought based on having similar thermal characteristics, such as loads, setpoints, schedules, exterior exposure, fenestration type and amount, and controls. This is further complicated in hospitals due to complicated ventilation requirements for positively pressurized and negatively pressurized zones, and zones that may require special air-handling systems. It is best to do a bottom-up analysis of these when creating a prototype model and describe what specific spaces are included in each "lumped" space type used in the model.

HOSP16-Zones (C1:C58)

The prototype includes the following named HVAC systems:

- VAV 1 – office, corridor, nurse's station, lobby
- VAV 2 – corridor, dining, imaging, nurse's station, physical therapy, office
- VAV ER – exam/treatment rooms
- VAV ICU – recovery room
- VAV Laboratory - labs
- VAV OR – operating room
- VAV Patient Room – patient room
- CAV Kitchen - kitchen

The quantity of HVAC systems seems much smaller than expected for a hospital model due to complicated ventilation requirements for positively pressurized and negatively pressurized zones, and zones that may require special air-handling systems. The nurses' station, in close proximity to patient rooms, has air that is recirculated in corridors, dining, and offices, which seems unlikely.

HOSP17-Zones (I1:I58)

A small analysis was performed, looking at the amount of area by space type for the hospital prototype as well as the AEDG 50% and Target 100 prototypes. For both the AEDG and Target 100, only about 85% of the spaces mapped well with the 13 space types used in the prototype. One of those space types, the office, was removed since the different prototypes either included medical office towers or did not. The fractions shown in the table below for AEDG and Target 100 are only the portions that could be mapped and exclude the office areas. The Target 100 prototype describes various "units" that were assumed to include both the nurses' stations and patient rooms.

Fractions of Area for Prototype, AEDG, Target 100

Space Type	Prototype	AEDG	Target100
Corridor Area	21.6%	13.6%	22.6%
Dining Area	3.9%	3.1%	1.5%
Exam/Treatment Room	2.2%	10.7%	6.4%
Imaging Room	2.7%	4.0%	7.2%
Kitchen/Food Prep Area	5.1%	1.4%	2.9%
Laboratory-General Health	2.9%	4.4%	3.1%
Lobby, Main Entry	8.2%	8.8%	2.0%
Nurse's Station	32.0%	8.0%	40.6%
Operating Room	3.4%	7.4%	8.0%
Patient Room	10.5%	34.7%	in nurses
Physical Therapy Room	2.7%	1.7%	2.9%
Recovery Room	4.9%	2.2%	2.7%

Several of the fractions in the prototype seem suspicious and are shown in yellow.

- Exam/Treatment Room, Imaging Rooms, and Operating Room fractions seem too small
- Kitchen/Food Prep Area and Recovery Room fractions seem too large
- Nurses Station/Patient Room total seems reasonable, but in the AEDG, the fractions are reversed for these two spaces.

The spreadsheet I used to perform this analysis is available by request and shows the mapping used.

HOSP18-Zones (J27:J42)

The number of people in the patient rooms is 102. I have assumed that this is the "number of beds," although that assumes that no patient rooms have visitors. Probably not the best assumption, but if a smaller number of beds is assumed, it further impacts the size of the hospital based on the number of beds, see comment HOSP9-Prototype.

HOSP19-Zones (n/a)

Another way of counting beds is based on the diagrams: from the Level 3 and Level 4 diagrams, there are 88 patient rooms on those floors, and on Level 2, another 12 intensive care unit patient rooms. That is a total of 100 patient rooms. That is close, but not exactly the number of patients, which is 102. That implies one room per bed, which is typical for modern hospital design.

HOSP20-Zones (J2:J57)

Based on the previous comment and using 102 beds, the number of other occupants in some of the space types seems inconsistent with this, always much higher than expected:

- Nurse's stations have 311 people, roughly three times the number of beds
- Corridors have 210 people, roughly twice the number of beds
- Lobbies have 524 people, roughly five times the number of beds

This means that either the area for these or the occupant densities for these space types is probably a poor assumption. It might be able to verify a better ratio of nurses to beds from one of the sources cited in comment HOSP3-Notes. The corridor and lobby numbers seem intuitively wrong based on my own visits to people in hospitals.

HOSP21-Envelope (A2:X22)

The main envelope table was reviewed, and only one error was found, see HOSP22-Envelope. A few rows were not reviewed: roof u-factor, windows u-factor, and SHGC for Pre-1978 and 1978-1997.

HOSP22-Envelope (M4, M9)

The exterior wall guest room and nonresidential for 1978-1997 vintage is shown as 0.256, which is based on Rt=3.9, but the value from Title 24-1986 shows Rt=4.6, which should be a U-value of 0.217. This column value was probably copied from the CZ6 value by accident.

HOSP23-Envelope (n/a)

Several envelope table issues were reported for the small hotel that also apply to the hospital prototype table:

- SHSC32 – roof replacement should be weighted
- SHSC32 – use of 0.338 instead of 0.33 as weighting
- SHSC35 – should say “11%-20%” not “10%-20%”
- SHSC37 – use non-north for all windows
- SHSC37 – consider weighting all standards for historic values
- SHSC39 – shouldn't average single and double pane window criteria since non-physical

For brevity, they will not be repeated.

HOSP24-Envelope (E39:X45)

To verify the infiltration values, the four and six side areas were based on the values from the Prototype tab:

- Area 4 sides = 56720 ft2
- Area 6 sides = 63203 ft2 + 2 x 40254 ft2 = 143711 ft2
- Ratio of 6 sides to 4 sides = 143711 / 56720 = 2.53

So, according to the referenced report

$$I_{design} \text{ at the building envelope } = 0.112 \times I_{75} \text{ at building envelope} \quad 2$$

$$I_{design\ (4-side)} = I_{design\ (5-or\ 6-side)} \times \left(\frac{\text{Total leakage area } (5 - \text{or } 6 - \text{side})}{\text{Exterior Wall area}}\right) \quad \textit{Equation 3}$$

Using these equations, the values should be:

Pre 1978	0.3117
1978-1997	0.3117
1998-2007	0.3117
2008-2018	0.2833
2019+	0.1700

These are very close to the values shown in the infiltration table; the differences are probably just due to rounding.

HOSP25-HVAC System (A2:H6)

While probably not required, the use of heat recovery is deployed in many hospitals since the heating and cooling requirements for hospitals, driven by large ventilation loads, are often high.

HOSP26-HVAC System (D17:F17)

In Title 24-2025, neither of the referenced tables 110.2 J nor 110.2-D shows boiler efficiencies, so I cannot verify the Ec = 85% used.

HOSP27-HVAC System (F16)

The Title 24-2013 table that contains the boiler efficiency is Table 110.2-K, not 110.2-J as shown.

HOSP28-HVAC System (D13:15, F13:F15)

The Title 24-2008 does not include a table concerning boiler efficiency. The reference to Table 112-G should be removed. Some boiler efficiency needs to be assumed, so it should be shown.

HOSP29-HVAC System (C13:C17)

The chiller efficiencies were all confirmed against their original source.

HOSP30-HVAC System (E13:E17)

The cooling tower efficiencies were all confirmed against their original source. The table reference for the cooling tower efficiencies should be added.

HOSP31-HVAC System (A25:I25)

Confirming fan power inputs with the PNL 20405 report

- Existing CAV TSP 4.09 – ok
- Existing FPFC TSP 1.33 – ok

The values from ACM 2025 shown in light red below were not confirmed. None of these values were found in ACM 2025. Please clarify exactly where these numbers are from and how they are derived.

			Fan power			
Vintage	Approach	System	Total Static Pressure (TSP) in w.c. [PNNL 20405, 2010, Table 5.11]	Fan Efficiency [ACM, 2025]	Fan Motor and Drive Efficiency [ACM 2025, Table 13]	W/cfm [ACM 2025, Table 14 and 15]
Existing	PNNL TSP	Kitchen System CAVMAU	4.09	0.65	0.9	
		All Other Zones-FPFC (on/off)	1.33			
Brake horse power (bhp)=(Supply cfm*TSP in w.c.)/(6356*Fan efficiency)						
Fan power (w/cfm) = bhp/motor and drive efficiency *746						
New Construction	ACM 2025	Kitchen System - VAVMAU System 6-VAV, >10,000 cfm	4.72	0.65	0.9	0.947
		All Other Zones System 6-VAV, >10,000 cfm	4.72			0.947

HOSP32-Thermostat (A2:S6)

Constant heating and cooling setpoints are appropriate for much of the hospital, but for offices and other spaces used primarily during the day, a setback at night is probably appropriate.

HOSP33-Thermostat (A2:S6)

Several special spaces should have unique thermostat settings.

- Operating rooms are often kept quite cool in the range of 60°F to 65°F
- Burn units are often kept much warmer than typical to make sure skin heals correctly and the patient's pain is reduced
- Imaging often depends on the requirements of the equipment and patients and are sometimes much cooler

This is not an exhaustive list; other special spaces may have different thermostat settings. In addition, some spaces in hospitals have special humidity requirements. See Table 4-A on the Ventilation tab. Some patient rooms may have an individually controlled thermostat to keep the occupants comfortable, and this implies a diversity of schedules for those rooms may be appropriate (see large hotel comment LHSC56-Thermostat).

HOSP34-Ventilation (O2:W16, A74:0115)

The table converting the ACH into a ventilation rate should be kept in the scorecard spreadsheet and not hidden or deleted in order to understand how the values are derived.

HOSP35-Ventilation (A74:0115)

Table 4-A shows the wide variety of special spaces in a hospital that need high outdoor ACH, high overall ACH, whether exhausted air is recirculated, design relative humidity, and design temperature. The large number of special spaces in a hospital is one of the driving reasons for the complexity and, typically, inefficiency of hospital HVAC systems, especially for existing hospitals.

HOSP36-Water Heater (n/a)

Missing space types for laundry and special sanitation, if present, would require additional water heaters to provide higher temperature water or perhaps boilers for providing steam.

HOSP37-Water Heater (A1:G3)

A building like a hospital that has boilers going all year to provide steam for sanitation purposes are likely to use the boiler with an indirect-fired water heater to provide hot water also. I don't think it is very common to have separate gas or electric water heaters.

HOSP38-Water Heater (A10:E22)

The hot water usage for different space types was checked against the source (App 5A of T24-2022) and was correct.

HOSP39-Water Heater (A33:B33)

The cold-water supply temperature of 55°F shown in the scorecard is consistent with the citation of the ACM 5.9.2 Rated Capacity, but better data is available that may be closer to how water heaters are actually sized using the local water mains temperature. The Appendix 5.4B (called CEC-400-2022-009-CMF-REV2-AP5.4B.xlsx) includes a WaterMain tab and WaterMainMonth tab that show climate-specific values for the entering water temperature. The lowest of these values for each climate zone is more likely to be used for sizing water heaters; the monthly values vary from 41.4 in CZ16 to 63.2 in CZ15. Using these values would have a large impact on the sizing of the water heater. This is the same comment as LHSC64-Water Heater and SHSC61-Water Heater.

HOSP40-Interior Lights (C4:C16)

The following space types allow additional lighting power, and it is very likely that for many of these space types, the additional lighting power would be used:

- Office Area (>250 square feet)
- Corridor Area
- Dining Area (Cafeteria/Fast Food)
- Healthcare Facility and Hospitals (Nurse's Station)
- Healthcare Facility and Hospitals (Recovery Room)
- Laboratory-General Health
- Healthcare Facility and Hospitals (Physical Therapy Room)
- Healthcare Facility and Hospitals (Imaging Room)
- Healthcare Facility and Hospitals (Patient Room)
- Lobby, Main Entry

A slightly different list applies to existing buildings, additional lighting power from T24-2019.

HOSP41-Interior Lights (B4:B16, E4:E16)

The lighting power densities for new construction (T24-2025) and existing (T24-2019) were checked and are ok.

HOSP42-Interior Lights (G4:G16)

No justification is used for using T24-2019 for existing lighting power densities. It seems a bit optimistic for many of the space types in a hospital to have their lighting systems replaced that quickly.

HOSP43-Ext Lights (E7:E13)

The lighting power allowances from Title 24-2025 were checked and are ok.

HOSP44-Ext Lights (n/a)

Where are the exterior lighting assumptions for the vintage models? One assumes that those should be higher. This is the same comment as LHSC72-Ext Lights and SHSC67-Ext Lights.

HOSP45-Ext Lights (A9:G9)

The parking area estimate states that it is based on "400 ft^2 for every parking space, one parking space per 830 ft^2 of gross building area," and the source is "2019 Outdoor Lighting CASE Report & PNNL 90.1-2010 TSD," but the value of 830 sqft from the CASE report is for warehouses:

Table 38: Assumptions for Converting Indoor New Construction Forecasts to Outdoor Spaces for General Hardscape – Table 140.7-A

Assumptions for Statewide Estimates - General Hardscape		
General Hardscape	**Assumptions**	**Area Multipliers for Construction S.F.**
for Large Office, Small Office, Food, Restaurant, College	1 parking space per 250 sf of gross building area	1
for Hotel, Retail, School, Other	1 parking space per 360 sf of gross building area	0.7
for NR Warehouse, Ref. Warehouse	1 parking space per 830 sf of gross building area	0.3

Furthermore, the areas shown in cell C9 of 386,198 sqft would not be correct based on that rule and instead would be 116,385 sqft.

A possible alternative way of computing this can be found here:

https://blog.spacemed.com/estimating-healthcare-facility-parking-requirements/

HOSP46-Ext Lights (A12:G12)

The building façade area assumes a factor of 0.5 of the perimeter walls; less than two stories are lit, but Table 36 from the CASE report suggests using 5% for "other" buildings that are not on the list.

HOSP47-Ext Lights (C11)

For hardscape ornamental lighting, the area should be 24,150, not 24,137 as shown. It is possible this difference is due to SI to IP rounding.

HOSP48-Ext Lights (B10:C10)

For the number of doors, this is usually based on fire code rules and is simplified in Table 36 of the CASE report to be 1 for each 5,000 sqft, which would be 48 entrances for the hospital, not 14 entrances or 10 entrances.

HOSP49-Ext Lights (B10:C10)

The non-sales canopy area shown of 4,827 sqft does not seem to correspond to anything in the CASE report, and no shading canopies are described anywhere else in the prototype scorecard. Please explain the source.

HOSP50-Equipment (A2:F15)

Several of the values shown for plug load power density are not consistent with the values from T24-2022 Appendix 5.4A, including office area, nurses' station, operating room, and imaging room. For the operating room and imaging room, the 5.4A values do not make sense, but it might be more appropriate to keep the office and nurses' station values consistent.

HOSP51-Equipment (A10:F10)

For operating rooms, 11 W/sqft is assumed. The value from T24-2022 Appendix 5.4A is 1.5 W/sqft, which is clearly too low. The AEDG 50% TSD uses 4 W/sqft (1 W/sqft plug load and 3 W/sqft process loads). The Target 100 analysis includes data on actually monitoring operating rooms.

HOSP52-Equipment (A12:F12)

For imaging rooms, 10 W/sqft is assumed. The value from T24-2022 Appendix 5.4A is 1.5 W/sqft, which is clearly too low. The AEDG 50% TSD uses 9 W/sqft (1 W/sqft plug load and 8 W/sqft process loads). The Target 100 analysis includes data on actually monitoring operating rooms.

HOSP53-Equipment (A19:B21)

Six elevators are assumed in the prototype, but that quantity is justified based on the office building estimation approach in the article:

https://buildingandinteriors.com/number-of-elevators-required-in-a-building/

The same article provides some guidance for medical buildings.

Another article uses similar guidance for medical facilities:

https://resoluteelevator.com/elevators-per-building/

This article implies that three elevators would be sufficient:

https://technoelevators.com/hospital-lifts-key-features-healthcare/

On the other hand, the Target 100 prototypes assumed 46 elevators for a 525,000 sqft building with 225 beds.

Overall, the quantity of elevator assumptions should be examined carefully.

HOSP54-Schedules (A1:AI27)

The schedules show zero occupancy at night. Hospitals do not have zero occupancy, let alone from 11 pm to 8 am on weekdays, 10 pm to 8 am on Saturdays, and 5 pm to 9 am on Sundays. A typical patient stay in a hospital is a few days to a week. The values taken from the Title 24-2022 file "CEC-400-2022-009-CMF-REV2-AP5.4B.xlsx" must describe a medical office building and are inappropriate for a hospital. The schedules for lighting, equipment, SHW, elevator, and gas equipment are all consistent with zero occupancy at night and should also be revised.

The AEDG 50% TSD has schedules in Appendix E for 24/7 areas, patient areas, office areas, food service-related areas, and laundry areas. The Targeting 100 work includes schedules in their Energy Modeling supplemental reports, described as eQUEST schedules.